

SLIDE-BY-SLIDE PRESENTATION COMPANION

Exactly What Every Slide Means, The Simple Idea, and What to Say Out Loud Without Getting Confused

Presenter: Beknan Chemedo | Project: Amharic Byte Mamba & Hayyuu AI

SLIDE 1: TITLE SLIDE: BYTE-LEVEL MAMBA VS. TRANSFORMER

👁️ **What the audience sees:** Dark navy screen with title, your name (Beknan Chemedo), and project topic.

💡 **Core Idea to Remember:** Introduce yourself with confidence and tell the audience that you built a faster, smarter Amharic AI that doesn't waste computer memory.

🍎 **Simple Analogy:** Think of this like introducing a new lightweight electric race car designed specifically for Ethiopian roads.

🗣️ **EXACT WORDS TO SAY OUT LOUD:**

"Good morning, everyone. My name is Beknan Chemedo. Today I am presenting my research on building a fast, lightweight, and token-free Generative AI for the Amharic language using Mamba State-Space Models and Brain-Inspired Continual Learning."

SLIDE 2: THE AMHARIC AI CHALLENGE: LOW-RESOURCE & TOKEN TAX

👁️ **What the audience sees:** 3 Cards showing: (1) Data Ceiling <500M words, (2) Token Tax (3.5x fertility), (3) Parameter Tax (63% wasted).

💡 **Core Idea to Remember:** Explain why normal AI struggles on Amharic: English has infinite data, but Amharic has very little clean data. Normal tokenizers cut Amharic words into confetti pieces, wasting 63% of the robot's brain.

🍎 **Simple Analogy:** Imagine trying to read a story where every single word is shredded into tiny puzzle pieces. You spend all your energy putting pieces together instead of understanding the story.

🗣️ **EXACT WORDS TO SAY OUT LOUD:**

"Current AI models like ChatGPT were designed for English. For Amharic, we face three big problems: First, we have less than 500 million clean words online. Second, standard tokenizers chop a single Amharic word into 7 or more tiny pieces—charging an unfair 3.5x 'Token Tax'. Third, a standard model wastes up to 63% of its entire brain capacity just storing a massive dictionary table instead of actual reasoning."

SLIDE 3: ARCHITECTURE DUALITY: QUADRATIC TRANSFORMER VS. LINEAR MAMBA

👁️ **What the audience sees:** Side-by-side comparison: Red box (Transformer $O(L^2)$) vs Green box

(Mamba SSM $O(L)$).

💡 **Core Idea to Remember:** Transformers look at every word and compare it to every other word (they get dizzy on long text). Mamba reads smoothly in a straight line, running 4x faster with constant memory.

🍎 **Simple Analogy:** Transformer is like a student who reads a paragraph by looking back and forth at every single word simultaneously. Mamba is like a reader who smoothly glides their finger along the line.

🗣️ **EXACT WORDS TO SAY OUT LOUD:**

"Because Amharic byte sequences are 3.5 times longer, normal Transformers become very slow because their self-attention compute explodes quadratically. In contrast, Mamba uses a Selective State Space Model that processes text in a straight linear line. It never gets bogged down, cuts latency by 75%, and uses 3 times less GPU memory."

SLIDE 4: EMPIRICAL BENCHMARK: TINYMAMBA (2.7M) OUTPERFORMS TRANSFORMER (13M)

👁️ **What the audience sees:** Figure 1 chart comparing model sizes vs Bits-per-Byte (BPB) compression scores.

💡 **Core Idea to Remember:** Our small 2.7M Mamba model beat a giant 13M Transformer! Remember: Lower BPB means a smarter model.

🍎 **Simple Analogy:** A small 2.7-liter lightweight sports car drove faster and used less fuel than a heavy 13-ton truck.

🗣️ **EXACT WORDS TO SAY OUT LOUD:**

"Here is our primary experimental result on 1.46 GB of clean Amharic text. We measured performance using Bits-per-Byte, where lower is better. Our TinyMamba model with only 2.7 million parameters achieved a score of 1.322 BPB, outperforming a 13 million parameter Transformer that scored 1.579 BPB. Our model was nearly 5 times smaller, but strictly more accurate!"

SLIDE 5: PARAMETER ALLOCATION TAX: RECLAIMING NEURAL CAPACITY

👁️ **What the audience sees:** Figure 2 bar chart showing how much of the model's brain is wasted on dictionaries.

💡 **Core Idea to Remember:** Tokenized models waste 63% of their weights on vocabulary lookup tables. By using raw bytes (256 numbers), we freed 98% of the model for pure thinking.

🍎 **Simple Analogy:** Buying a 100-page notebook where 63 pages are blank index tables vs a notebook where 98 pages are full of brilliant thinking.

🗣️ **EXACT WORDS TO SAY OUT LOUD:**

"Why did our smaller model win? Because of the Parameter Allocation Tax shown in this chart. The 32k Transformer wasted 62.8% of all its weights on a static vocabulary dictionary. By moving to raw UTF-8 bytes with only 256 entries, our model dedicated 97.7% of its parameters to deep neural

reasoning layers."

SLIDE 6: CATASTROPHIC FORGETTING & COMPLEMENTARY LEARNING SYSTEMS

👁️ **What the audience sees:** Red box (Standard Backprop forgetting) vs Green box (Brain-Inspired Dual Memory).

💡 **Core Idea to Remember:** Standard AI forgets past grammar when learning today's news. We copied the human brain: a fast notepad for today's facts + a slow permanent brain for language grammar.

🍎 **Simple Analogy:** When you meet someone new, you write their phone number on a sticky note without forgetting how to speak your native language.

🗣️ **EXACT WORDS TO SAY OUT LOUD:**

"When you update standard AI on live news, it suffers from Catastrophic Forgetting—grammar retention collapses down to 42%. To solve this, we drew inspiration from human neuroscience: the Complementary Learning Systems theory. We gave our AI two cooperating memory systems: a fast Hippocampus for instant fact capture, and a slow Mamba Neocortex for stable grammar."

SLIDE 7: HEBBIAN FAST WEIGHTS & ANTI-HOPFIELD SYNAPTIC ENGRAMS

👁️ **What the audience sees:** Figure 5 diagram showing the dual-memory neural flow and 1-shot Hebbian matrix.

💡 **Core Idea to Remember:** New facts are written into a special memory matrix (M) in 0.01 seconds without running slow backpropagation.

🍎 **Simple Analogy:** Taking a instant Polaroid photo of a fact and pinning it to a corkboard in 0.01 seconds.

🗣️ **EXACT WORDS TO SAY OUT LOUD:**

"When a user teaches Hayyuu a new fact, it writes the information into a Hebbian fast weight matrix in just 0.01 seconds using an anti-Hopfield outer-product update. This completely avoids gradient updates to the main model, achieving 100% immediate factual recall with zero linguistic interference."

SLIDE 8: OFFLINE SLEEP CONSOLIDATION: SHARP-WAVE RIPPLE REPLAY

👁️ **What the audience sees:** Figure 3 chart showing grammar retention staying at 100% after sleep consolidation.

💡 **Core Idea to Remember:** Every 30 rounds, the AI takes a sleep break and replays daytime memories 20x faster into its permanent neural weights.

🍎 **Simple Analogy:** Studying flashcards during the day, and while you sleep at night, your brain

moves those memories into long-term storage.

 EXACT WORDS TO SAY OUT LOUD:

"Just like biological brains, our model enters an offline sleep cycle every 30 rounds. During sleep, it replays its daytime memories at 20 times speed alongside foundational anchor buffers. As you can see in Figure 3, this permanently bakes the new facts into the Mamba weights while maintaining 100% grammar retention."

SLIDE 9: HAYYUU: AUTONOMOUS LIVING CONVERSATIONAL AGENT

 **What the audience sees:** Figure 4 SFT convergence curve and Telegram bot deployment overview.

 **Core Idea to Remember:** We fine-tuned the model on 4,359 high-quality Amharic Q&A pairs (dropping loss to 0.695 BPB) and launched Hayyuu live on Telegram.

 **Simple Analogy:** Taking our smart student through graduation school so it learns how to politely answer questions on Ethiopian history, culture, and science.

 EXACT WORDS TO SAY OUT LOUD:

"We aligned our pre-trained model into a helpful conversational assistant named Hayyuu by fine-tuning on 4,359 native Amharic instruction pairs. Figure 4 shows the rapid SFT convergence down to 0.695 BPB. We then connected Hayyuu to Telegram with dual-candidate RLHF feedback and an autonomous 24/7 self-teaching teacher."

SLIDE 10: LIVE DEMONSTRATION: AMHARIC REASONING & IDENTITY

 **What the audience sees:** 2 Cards with real Amharic conversation prompts on History, AI, Culture, and Identity.

 **Core Idea to Remember:** Show real examples of Hayyuu answering questions accurately: It knows who created it (Beknan Chemed), where Axum is, and what AI means in pure Amharic.

 **Simple Analogy:** Showing the robot speaking fluent Amharic in front of the classroom.

 EXACT WORDS TO SAY OUT LOUD:

"Here are sample outputs from Hayyuu. When asked 'Who created you?', it correctly identifies its creator, Beknan Chemed. When asked about Ethiopian history like the Obelisk of Axum or modern science like Artificial Intelligence, it generates fluent, grammatically accurate Amharic responses."

SLIDE 11: SUMMARY OF CONTRIBUTIONS & GLOBAL IMPACT

 **What the audience sees:** 3 Green summary cards: (1) Token-Free Efficiency, (2) Zero-Forgetting CLS, (3) Sovereign Ethiopian AI.

 **Core Idea to Remember:** Wrap up with the 3 big wins: No token tax, no forgetting, and proving

that African AI can run on a single affordable GPU.

🍎 **Simple Analogy:** Proving that smart engineering beats expensive supercomputers.

🗣️ **EXACT WORDS TO SAY OUT LOUD:**

"In conclusion, this research makes three major contributions: First, we eliminated the Token Tax for African languages using byte-level Mamba. Second, we solved catastrophic forgetting using a brain-inspired dual-memory architecture. Third, we proved that sovereign, high-performance generative AI can be trained on a single affordable graphics card right here in Ethiopia."

SLIDE 12: CLOSING / Q&A DEFENSE

👁️ **What the audience sees:** Dark navy closing screen with 'THANK YOU!', Questions & Discussion, and repository links.

💡 **Core Idea to Remember:** Thank the audience with a smile and invite questions. You know all the answers!

🍎 **Simple Analogy:** Crossing the finish line after a perfect presentation.

🗣️ **EXACT WORDS TO SAY OUT LOUD:**

"Thank you very much for your time and attention. I am now happy to answer any questions!"